

AT82.05 Natural Language Processing

Team StochasticallyYours presents >

PersonaVerse: Persona Driven Survey Synthetic Response Generation

"All the world's a stage, And all the agents merely players"

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**ALL THE
WORLD'S
A STAGE**

...



**...& ALL
AGENTS,
MERELY
PLAYERS**

Scaling Synthetic Data Creation with 1,000,000,000 Personas

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Generative Agent Simulations of 1,000 People

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Abstract:

The promise of human behavioral simulation—general-purpose computational agents that replicate human behavior across domains—could enable broad applications in policymaking and social science. We present a novel agent architecture that simulates the attitudes and behaviors of 1,052 real individuals—applying large language models to qualitative interviews about their lives, then measuring how well these agents replicate the attitudes and behaviors of the individuals that they represent. The generative agents replicate participants' responses on the General Social Survey 85% as accurately as participants replicate their own answers two weeks later, and perform comparably in predicting personality traits and outcomes in experimental replications. Our architecture reduces accuracy biases across racial and ideological groups compared to agents given demographic descriptions. This work provides a foundation for new tools that can help investigate individual and collective behavior.

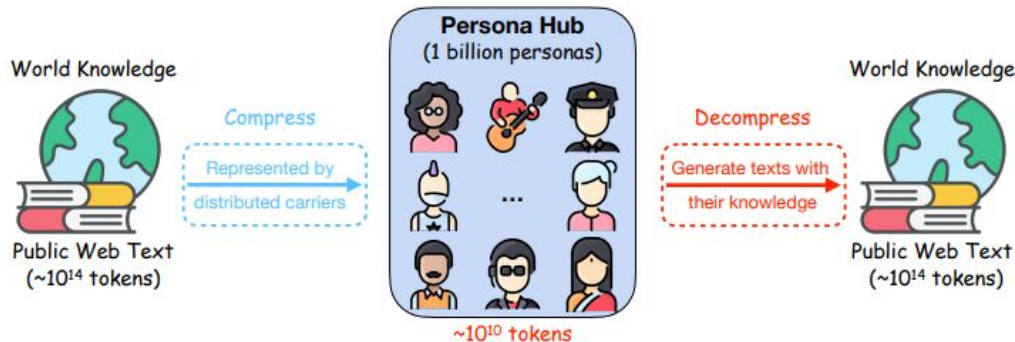
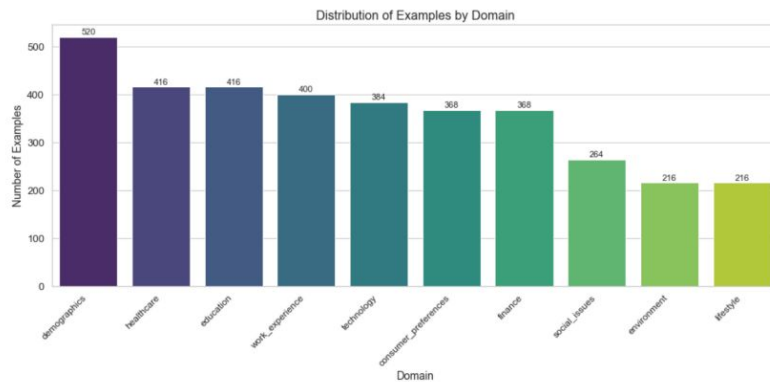


Figure 2: From a compression perspective (Delétang et al., 2023; Ge et al., 2024), Persona Hub ($\sim 10^{10}$ tokens) can be seen as the compressed form of world knowledge (public web text for training LLMs, $\sim 10^{14}$ tokens) into distributed carriers. On the other hand, the public web text can be seen as the decompressed content created by these personas with their knowledge and experiences.

We build our dataset on top of the PersonaHub Dataset provided by Tencent AI Lab

```
{
  "id": "@01199",
  "domain": "consumer_preferences",
  "persona": "A young marketing professional looking to learn the art of product positioning and brand building from the expert",
  "question": "How well did the product/service you purchased meet your overall expectations?",
  "system_prompt": "You are embodying the following persona: A young marketing professional looking to learn the art of product positioning and brand building from the expert\n\nPlease respond to survey questions authentically as this persona would. Consider the persona's background, values, priorities, and likely perspectives while answering. Provide thoughtful, realistic responses that reflect the persona's authentic point of view. Keep responses conversational but substantial, typically 3-5 sentences.",
  "user_prompt": "Survey Question: How well did the product/service you purchased meet your overall expectations?\n\nPlease provide your honest and detailed response to this question.",
  "response": "I recently purchased a new productivity app that was marketed as a game-changer for managing projects and collaborating with teams. Honestly, it met my expectations pretty well! The user interface is intuitive, which makes onboarding new team members a breeze, and I love the integration options it offers with other tools we already use. However, I did notice that the mobile version is a bit limited in functionality compared to the desktop version, which was slightly disappointing since I often work on the go. Overall though, it's been a solid addition to our toolkit and has definitely helped streamline our workflow.",
  "response_length": 623,
  "persona_length": 113,
  "question_length": 78,
  "question_sentences": 1,
  "estimated_tokens": 155.75,
  "domain_persona": "consumer_preferences:A young marketing professional looking to learn the art of product positioning and brand building from the expert"
}
```

Domain	Training	Validation	Test
consumer_preferences	294	38	36
demographics	416	59	45
education	333	42	41
environment	173	18	25
finance	294	36	38
healthcare	333	43	40
lifestyle	173	17	26
social_issues	211	27	26
technology	307	38	39
work_experience	320	39	41



- Training set: 2854 examples
- Validation set: 357 examples
- Test set: 357 examples
- Total: 3568 examples

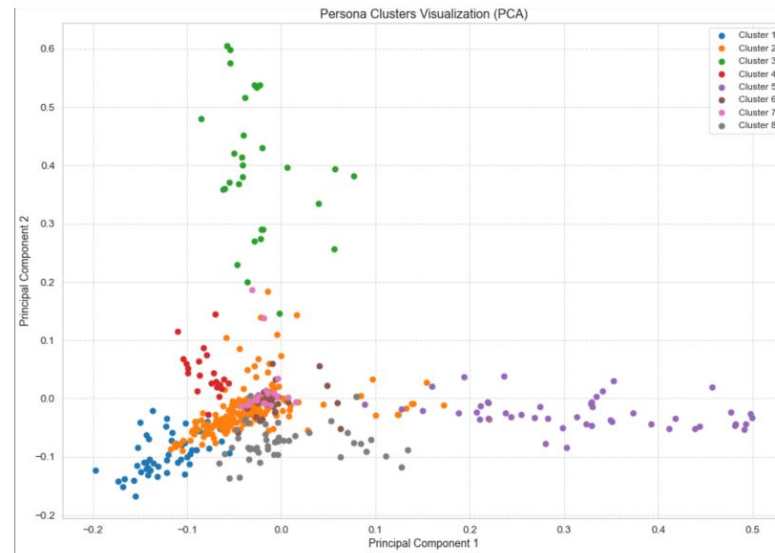
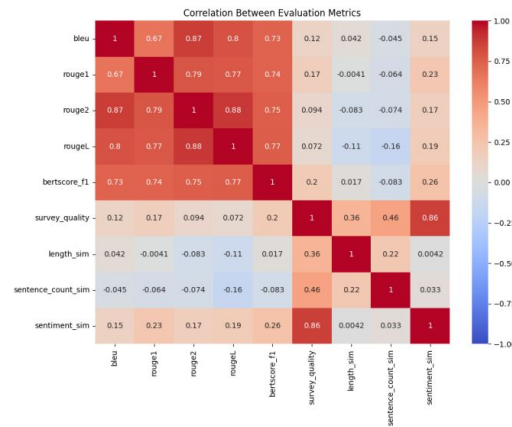
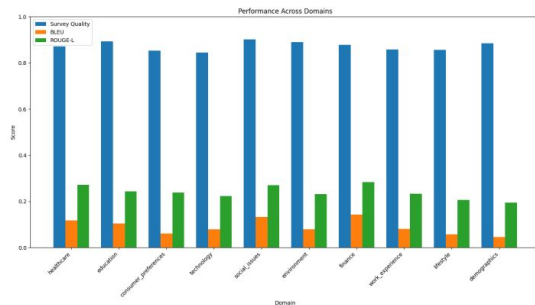
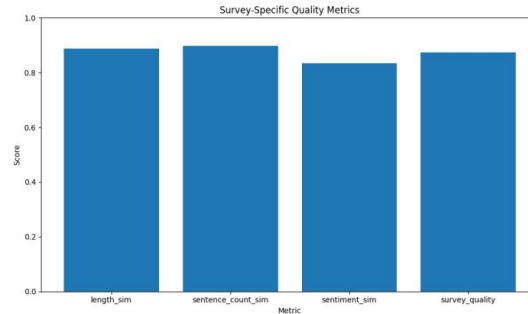
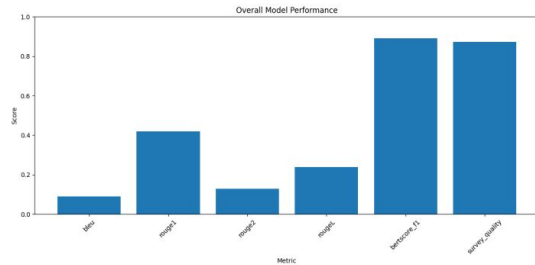


Figure 3: PersonaVerse Distribution of examples per domain(left) and the Persona Clusters(right)

Base Language Model	Reason	Embedding Models	Reason
TinyLlama 1.1B	Lightweight architecture (1.1B params); Optimized for instruction tuning; Highest BLEU score despite small size	FastText	Efficient subword embeddings for survey text tokenization
Microsoft Phi-2	Strong reasoning (2.7B params); Trained on synthetic data; Highest ROUGE1 scores and excellent quality	BERT-Mini	Low-latency embeddings compatible with Phi-2's architecture
Qwen2 1.5B	Bilingual support; Efficient parameter usage at 1.5B; Strong sentiment preservation	all-MiniLM-L6-v2	Optimized for semantic similarity in short responses
Llama 3.1 8B Lexi	Latest LLaMA iteration; 8B parameter size balances capacity/efficiency; Uncensored for wider response range	Sentence Transformers	Versatile framework with strong semantic understanding
Mistral 7B	Consistent output quality; Strong structural similarity; Excellent sentence count preservation across domains	USE	Fast inference, fine-tuned for question-answering tasks
DeepSeek R1 Distill Qwen2 1.5B	Knowledge distillation from larger models; Top survey quality (0.8817); Best sentiment similarity	MPNet	Strong contextual embeddings for sentiment alignment
Qwen1.5 MoE	Mixture-of-Experts architecture; Specialized pattern capabilities; Good text metrics despite structural limits	E5-small	Efficient embeddings designed for retrieval tasks

Table 3: Base Language Models and Embedding Models Used in Survey Response Generation

 aryashah00/survey-finetuned-tinyllama-for-dep... Text Generation • Updated 3 days ago • ⬇ 6
 aryashah00/survey-finetuned-Mistral-7B-Instru... Text Generation • Updated 8 days ago • ⬇ 8
 aryashah00/survey-finetuned-DeepSeek-R1-Disti... Text Generation • Updated 8 days ago • ⬇ 8
 aryashah00/survey-finetuned-phi-2 Text Generation • Updated 16 days ago • ⬇ 3
 aryashah00/survey-finetuned-Qwen1.5-MoE-A2.7B Text Generation • Updated 7 days ago • ⬇ 5
 aryashah00/survey-finetuned-Llama-3.1-8B-Lexi... Text Generation • Updated 8 days ago • ⬇ 10
 aryashah00/survey-finetuned-Qwen2-1.5B-Instru... Text Generation • Updated 16 days ago • ⬇ 5
 aryashah00/survey-finetuned-TinyLlama-1.1B-Ch... Text Generation • Updated 16 days ago • ⬇ 99 • ❤ 2



Model	BLEU	ROUGE1	BScore	SQ	SS
DeepSeek R1	0.071	0.404	0.884	0.882	0.870
Llama 3.1 8B	0.053	0.392	0.883	0.805	0.803
Mistral 7B	0.078	0.416	0.889	0.867	0.800
Phi-2	0.087	0.429	0.891	0.868	0.820
Qwen1.5 MoE	0.050	0.356	0.854	0.574	0.788
Qwen2 1.5B	0.053	0.377	0.883	0.866	0.845
TinyLlama 1.1B	0.090	0.419	0.890	0.873	0.835

Table 4: Overall performance comparison across models. BScore = BERTScore F1, SQ = Survey Quality, SS = Sentiment Similarity.

Model	BLEU	ROUGE1	SQ	SS
DeepSeek R1	0.027	0.024	0.033	0.097
Llama 3.1 8B	0.023	0.025	0.037	0.073
Mistral 7B	0.029	0.028	0.031	0.077
Phi-2	0.028	0.030	0.035	0.078
Qwen1.5 MoE	0.022	0.021	0.042	0.112
Qwen2 1.5B	0.021	0.025	0.024	0.056
TinyLlama 1.1B	0.035	0.019	0.019	0.066

Table 5: Standard deviation of performance across domains. SQ = Survey Quality, SS = Sentiment Similarity. Lower values indicate more consistent performance.

Figure 4: TinyLlama 1.1B Performance Metrics

Key Performers:

- **TinyLlama 1.1B:** Highest BLEU (0.090)
- **Phi-2:** Highest ROUGE1 (0.429) & BERTScore (0.891)
- **DeepSeek R1:** Best sentiment preservation (0.870)
- **Mistral 7B:** Best structural consistency (0.913)

Strongest Domains:

- **Social Issues:** TinyLlama (BLEU: 0.132), Phi-2 (ROUGE1: 0.477)
- **Finance:** TinyLlama (BLEU: 0.142), Mistral 7B (BLEU: 0.139)
- **Technology:** Phi-2 (Survey Quality: 0.936)

Most Challenging Domains:

- Consumer Preferences
- Lifestyle

Surprise Finding: Small Models Win

TinyLlama 1.1B (1.1B parameters)

- Highest BLEU score (0.090)
- Strong across all metrics
- Excellent parameter efficiency

Phi-2 (2.7B parameters)

- Highest ROUGE1 score (0.429)
- Best semantic similarity (BERTScore: 0.891)
- Strong performance in technology domain

Significantly outperforming larger models!

Most Consistent Models:

- **Qwen2 1.5B:** Lowest variation across domains
- **TinyLlama 1.1B:** Most consistent survey quality

Inconsistent Models:

- **Qwen1.5 MoE:** High variation in sentiment (SD: 0.112)

TinyLlama 1.1B: Best n-gram precision, excellent efficiency

Phi-2: Best semantic understanding, strong in technology domain

DeepSeek R1: Best sentiment preservation, highest survey quality

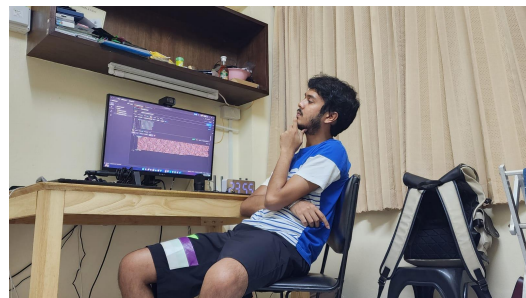
Mistral 7B: Most balanced performance, best structural consistency

Qwen2 1.5B: Most consistent cross-domain performance

But Wait...!



Thank You :)





One last time ;)

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